

IMAGE PROCESSING · FINAL PROJECT

Image Processing Project

Feature-Based Image Processing and Reinforcement Learning for CarRacing-v3



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16-D base observation



PPO 938.87 @ 500K



SAC 938.51 @ 400K

MOTIVATION

From raw pixels to a compact, inspectable representation

THE PROBLEM

Training a policy directly on the 96×96×3 pixel frame couples perception and control. It is sample-hungry, opaque, and expensive to debug on a course budget.

THIS PROJECT

Move perception into a deterministic image-processing wrapper. Off-the-shelf PPO and SAC then learn only control on a compact 16-D feature vector.

27,648 pixels → **16** values

Each frame becomes a vector an MLP can act on —
four frames stack to 64-D.

ENVIRONMENT

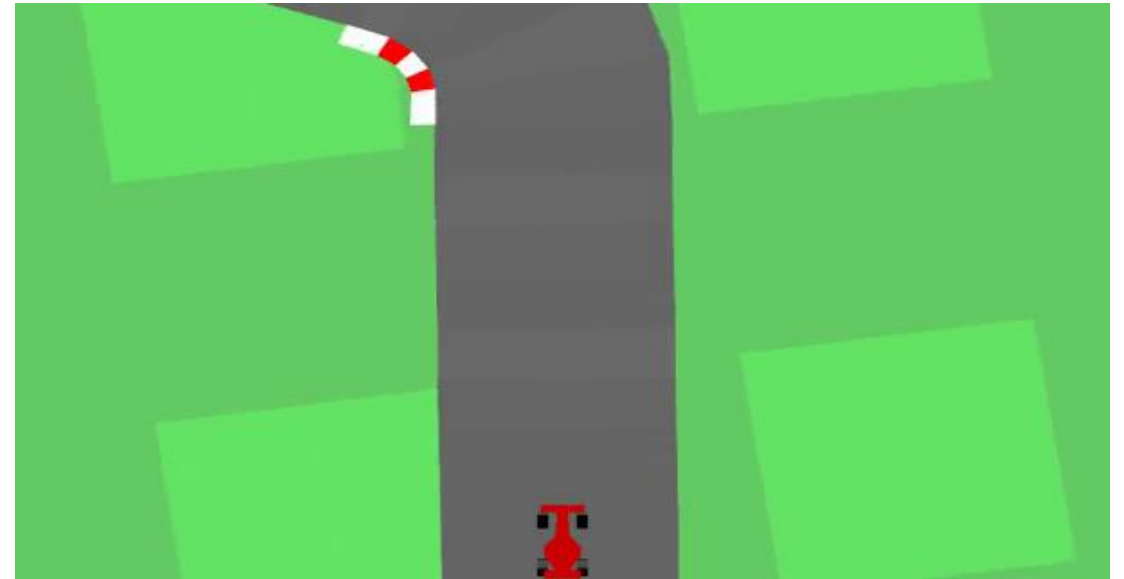
Gymnasium CarRacing-v3

96×96 RGB frame — top-down view around the car.

Continuous action [steer $\in -1...1$, gas, brake].

Reward = progress along the track; penalties off-track.

Randomized tracks each episode $\rightarrow$ must generalize, not memorize.



Top-down rendered driving frame (road, grass, car).

Two ways to drive from vision

A

End-to-end pixels (CNN)

- Learns perception and control jointly.
- General, but sample-hungry.
- Opaque and hard to debug.
- Heavy compute footprint.

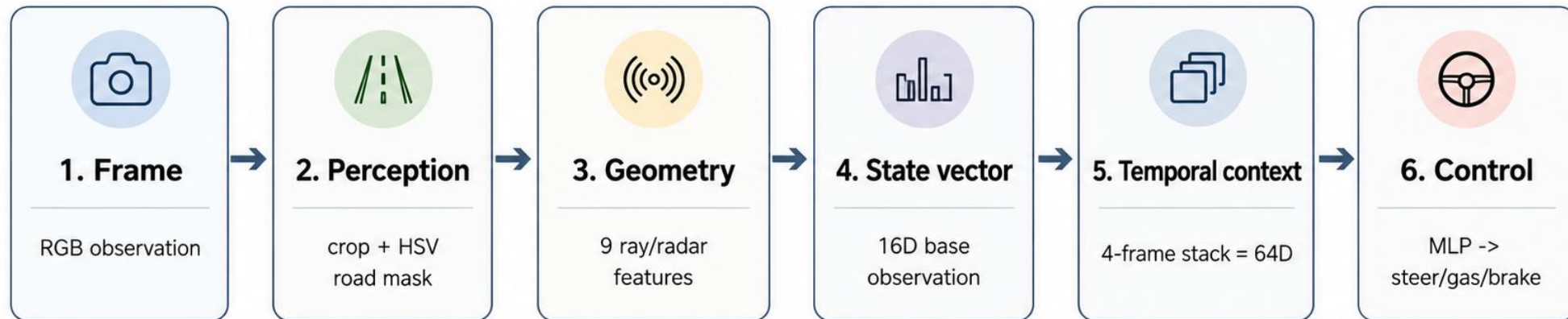
B

Feature-based (this work)

- Deterministic CV → compact geometry.
- Sample-efficient and inspectable.
- MLP-sized policy input (16-D / 64-D).
- Every stage is auditable.

No raw-pixel CNN policy is trained here — the contribution is the perception layer, not a new learner.

The image-processing pipeline



Observation composition: 9 rays + 3 smoothed actions + speed + curvature + front-ray change + left-right asymmetry



This is feature-based visual control; the policy is not trained directly on raw image tensors.

HSV road-mask extraction

1

Crop + RGB→HSV

Keep the driving region; HSV separates color from brightness.

2

Threshold grass → complement

Select the green band, invert to get the road.

3

3×3 morphological opening

Erode then dilate — removes speckle and pin-holes.

4

Connected components

Keep the road blob at the car; drop stray patches.

HSV GRASS BAND

$[28, 35, 35]$
– $[92, 255, 255]$

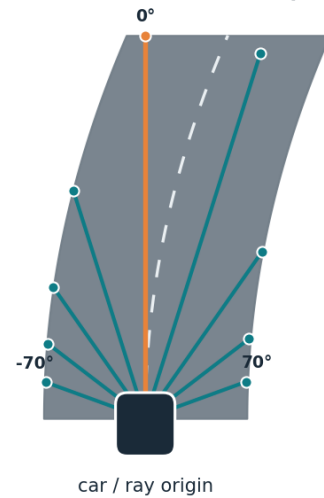
$H \in 0-179, S, V \in 0-255$

road = ¬ grass

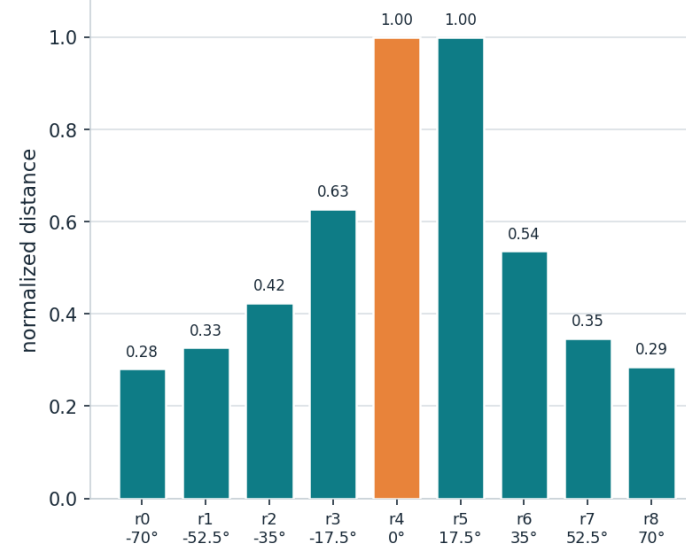
Selecting the uniform grass and inverting is more stable than thresholding the road directly.

Ray / radar feature extraction

(a) Nine-ray fan on the road mask ($-70^\circ \dots +70^\circ$)

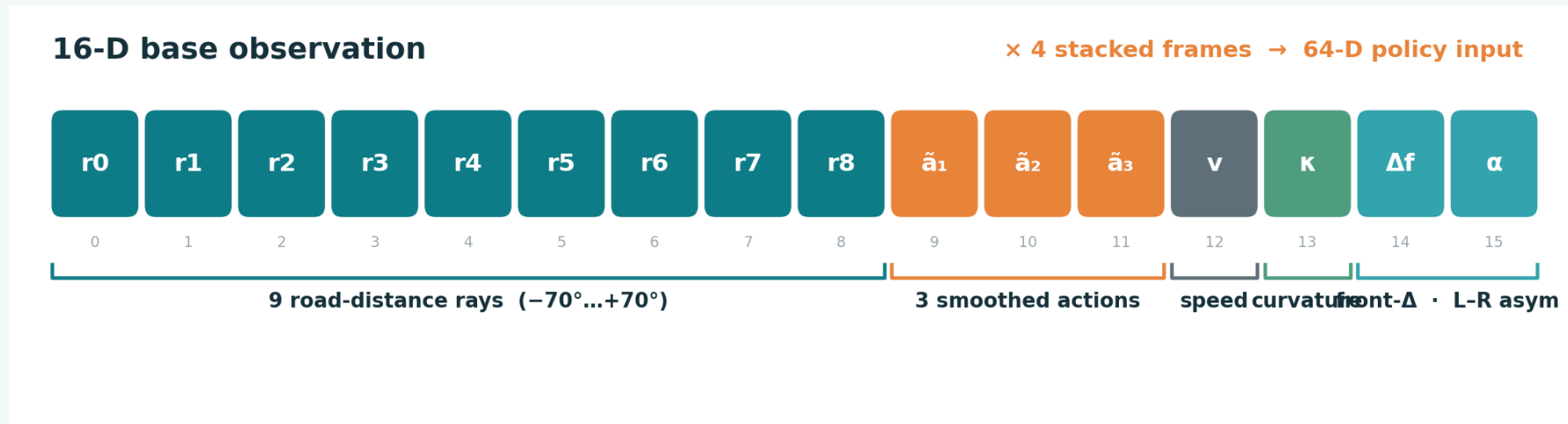


(b) Nine normalized ray distances $\rightarrow r_0 \dots r_8$



Nine rays span $-70^\circ \dots +70^\circ$, march to the mask boundary, and return normalized distances $r_0 \dots r_8$ — a compact profile of local road openness.

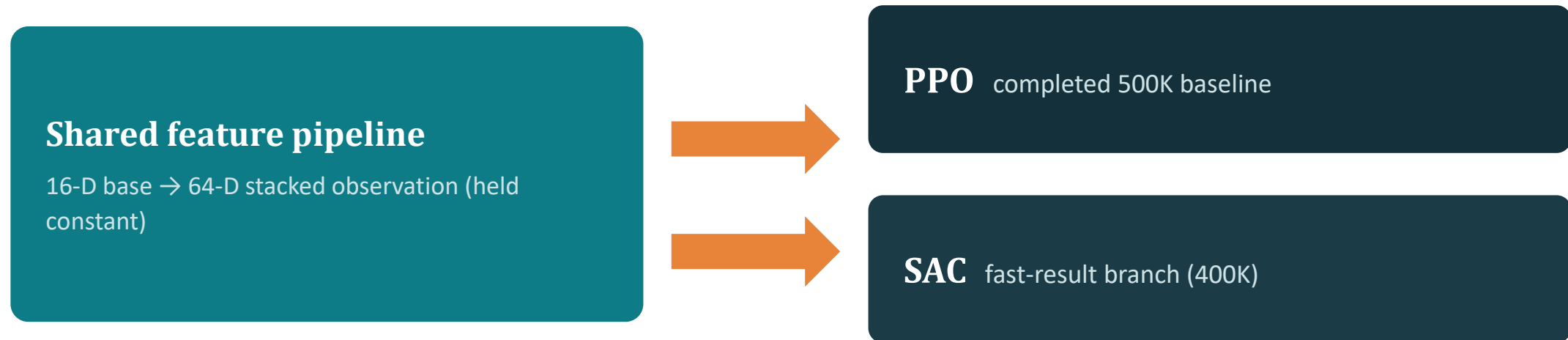
16-D base → 64-D stacked observation



16-D = 9 ray distances + 3 smoothed actions + speed + curvature + front-ray change + left–right asymmetry.

Four consecutive observations stack (×4 → 64-D) to capture short-horizon motion. PPO and SAC share this exact stream.

One perception layer, two off-the-shelf learners



Perception is held constant; only the policy optimizer changes. **Any behavioral difference is attributable to the learner, not the representation.**

Both are unmodified Stable-Baselines3 algorithms with MLP policies.

RESULT · PPO

PPO completed baseline

VALIDATED @ 500,000 STEPS

938.87 ± 7.86

Best parsed **939.53 ± 4.09** @ 480,000

- ✓ Fully completed 500K run (last step 501,760).
- ✓ ~40 fps · ~3 h 26 m on Colab.
- ✓ Stabilizes in the low-930s–940s band over final checkpoints.

Raw reward (track progress) is the headline metric; shaped reward is a training aid only.

RESULT • SAC

SAC fast-result branch

VALIDATED @ 400,000 STEPS • PARTIAL

938.51 $\pm$ 4.88

Best *and last* eval checkpoint — none after 400K.

- Not a completed 500K run (last step 418,697 $\approx$ 82%).
- Fast-result branch under Colab runtime limits.
- **Not claimed to beat PPO; not compute-equivalent.**

Treated strictly as a 400K best-checkpoint result — the conservative reading of the evidence.

PRIMARY RESULTS

Headline raw-reward comparison

PPO @ 500K

COMPLETED 500K

● **938.87** ± 7.86

PPO best @ 480K

BEST PARSED

● **939.53** ± 4.09

SAC @ 400K

PARTIAL 400K

● **938.51** ± 4.88



Raw reward is the headline metric. PPO and SAC are not a compute-equivalent comparison.

Raw eval reward vs training steps (from log checkpoints).

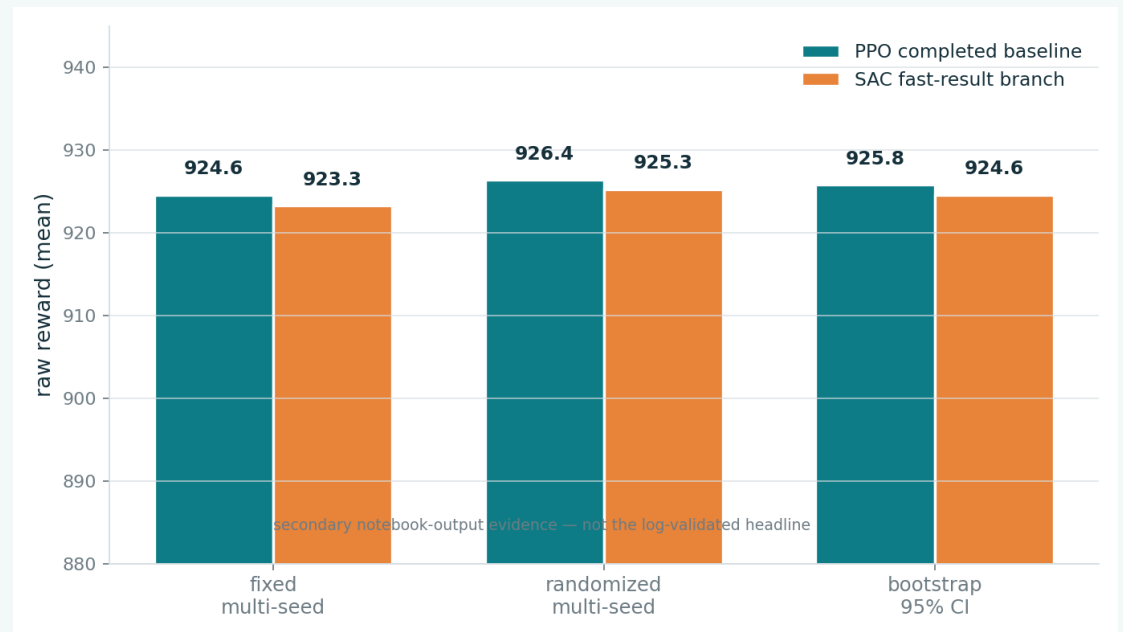
Generalization across seed sets

Fixed and **randomized** multi-seed sets, plus a bootstrap 95% CI.

Both branches land in the mid-920s on held-out seeds — a few points below the best single checkpoints.

Narrow confidence intervals → the mean behavior is stable even as individual seeds vary.

Secondary notebook-output evidence — distinct from the log-validated headline metrics.



Limitations



SAC is partial

400K best checkpoint — not 500K, not compute-equivalent, not claimed to beat PPO.



No pixel baseline

No raw-image CNN policy was trained, so there is no head-to-head pixel comparison.



Appearance-sensitive

HSV masking can degrade under rendering or track-art changes.



Heavy-tailed failures

Qualitative limitation clips
Some high-reward episodes still show unstable recovery, brief off-road drift, or sharp-turn weakness.



Not “solved”

CarRacing-v3 is not claimed solved; promising, not uniformly robust.

**Reported as multi-seed statistics —
never a single best episode.**

TAKEAWAYS

A compact visual feature that two RL methods can drive on

- 1 **Perception is the contribution** — a deterministic HSV-mask + ray pipeline turns pixels into a 16-D / 64-D feature.
- 2 **Shared, learner-agnostic** — PPO (938.87 @500K) and SAC (938.51 @400K) reach the same band on the same features.
- 3 **Honest and reproducible** — conservative claims, log-parsed metrics, saved artifacts — no retraining to verify.

GitHub

<https://github.com/UltraLamb/Image-Processing-Project>

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